JOSHI NINAD DILIPRAO

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.55
SBES College of Science	Class XII (Maharashtra State Board)	2020	78.77%
Mukul Mandir	Class X (Maharashtra State Board)	2018	91.00%

POSITIONS OF RESPONSIBILITY

TedX - Operations and Logistics(Sub-coordinator)

Finance and Consulting Club - Design (Coordinator)

Office of Career Services - Placement Coordinator

Tinkerer's Lab - Operations and Logistics(Deputy Coordinator)

PROJECTS

Design and Fabrication of an Underwater Glider

- Developed a **servo-actuated underwater glider** for energy-efficient, long-distance travel.
- Designed a **syringe-based** mechanism to **control** water intake and expulsion.
- Varied **buoyancy** to generate forward motion through weight, lift, and drag.

Simultaneous Localization and Mapping(SLAM) of 4 wheel drive vehicle

- Simulated a four-wheel-drive vehicle using **ROS 2** plugins and the **Nav2** navigation stack.
- Implemented **SLAM-based navigation** to evaluate localization and mapping.
- Deployed the architecture on a **3D-printed robot** with wheel encoders, **LiDAR**, and BLDC motors.

Development of Manual Road Cleaning Device

- **Designed and fabricated** a fully mechanical road cleaning device.
- Engineered a **gear-driven transmission** to convert wheel motion into broom rotation.
- Enabled **debris sweeping** through manual pushing without external power.

Breast Cancer Classification Using Machine Learning

- Compared **SVM, KNN, and Random Forest** models for breast cancer classification.
- Applied **feature standardization** and **stratified train-test splitting**.
- Evaluated **accuracy, precision, recall, and F1**, achieving **97.9% SVM accuracy**.

Handbag and Shoe Classification

- Implemented **CNN** and **Vision Transformer** models for handbag and shoe classification.
- Applied **image augmentation** and adapted an **ImageNet-pretrained** Vision Transformer.
- Achieved **90.0%** CNN test accuracy and compared results with Transformer models.

Lyrics-Based Music Genre Classification

- Developed **neural-network classifiers** to predict music genres from lyrics.
- Trained on **48,991 examples** with separate validation and test datasets.
- Compared model configurations, achieving **74.1% accuracy** on **21,774 test examples**.

California Housing Value Prediction

- Built an **XGBoost** regression model using **20,640** California census block groups.
- Performed **exploratory and correlation** analyses to examine housing-value predictors.
- Achieved a test **R^2 of 0.834** on **4,128** held-out samples.

Facial Expression Recognition Using Deep Learning

- Developed **CNN classifiers** for seven expressions using **35,887** FER2013 images.
- Compared CNN architectures, **image augmentation**, and VGG19 **transfer learning**.
- Improved test accuracy from **47.4% to 54.3%** with augmentation and a wider CNN.

Transformer-Based Slot Filling for Flight Queries

- Implemented a **Transformer encoder** with positional embeddings and multi-head attention.
- Built **text preprocessing and sequence-labeling pipelines** for the ATIS dataset.
- Achieved **96.4%** test token accuracy excluding padding on flight queries.

Design, Development and Demonstration of Tilt-Rotor Quadrotor System

- Developed a **mathematical model** for tilt-rotor quadrotor control and simulation.
- Designed **PID and neural-network controllers** to regulate system response.
- Investigating **reinforcement learning** to improve tilt-rotor control adaptability.

Optimizing Battery Cooling in Electric and Hybrid Vehicles

- Developed a **2D EV thermal model** using **CFD** to assess cooling performance.
- Evaluated **fan positions** to improve airflow and thermal management.
- Optimized **fan speed and placement** to balance cooling and power consumption.

Design and Fabrication of Front and Rear wing for SAE Vehicle

- Compared **carbon-fiber and glass-fiber composites** for SAE vehicle wings.
- Studied **composite fabrication processes** to inform material selection.
- Analyzed how **airfoil angle of attack** influences aerodynamic forces and vehicle dynamics.

INTERNSHIP

ITC Ltd.

- Reduced laminate wastage on Aashirvaad Atta Manufacturing line from 5% to 2%.
- Prepared Piping and Instrumentation Diagram of complete plant
- Developed and implemented Micro-checklist for 5S practices.

HOBBIES AND INTERESTS

Badminton, Swimming, Reading Books, Riding Bicycle, Hindu Scriptures and stories

CERTIFICATIONS

Google Advanced Data Analytics Professional Certificate

Google Project Management Professional Certificate